

Matlab Assignment #3

Q1) For the following function

$$f(x, y) = 1 - [\exp(-(x^2 + y^2)) + (x^2 + y^2)/20],$$

1. Plot $f(x, y)$ using surface plot.
2. Find the global minimum for the variables x and y .

Q2) Solve the following system of differential equations and plot the results of the two solutions on the same figure.

$$\begin{aligned} \frac{dy_1}{dx} &= 2y_1 + 3y_2 \\ \frac{dy_2}{dx} &= -5y_1 + 6y_2 \end{aligned}, \quad \text{where } y_1(0) = 0, \quad y_2(0) = 1$$

Note the following:

- The reference for the above material is available in Part3.
- Prepare your file in a pdf format and submit it before the due date.
- The maximum number of students per assignment is two.
- The pdf file should contain your names, Matlab scripts, results, plots, and conclusions.
- Share Matlab scripts among other groups is not allowed.